

The problem

Bellcourt Health Administrators | prior authorization | Lohith Devaramane

Bellcourt runs health plans for **38 employers** and decides prior authorization requests for an insurer, **Riverbend**, which audits it, penalises it, and can leave on 31 December.

Prior authorization: is the service covered, and is it medically necessary? Nurses approve. Only physicians deny. Deadlines are legal: 7 days, 15 days, 72 hours urgent. About 310,000 requests a year, 46% by fax.

90.1%

Decisions on time

Standard 97%

56.7%

Correct in QA audit

Standard 95%

56.2%

Denials overturned on appeal

Trigger 30%

\$1.9M

Penalties this year

A fifth of profit

Leadership's explanation: **"It's a staffing problem, full stop."** Twenty more nurses. The CEO is not convinced.

How I found it

Every stakeholder claim tested against the data pack: 2,400 requests, 120 audited cases, 90 weeks of staffing, 3,200 calls

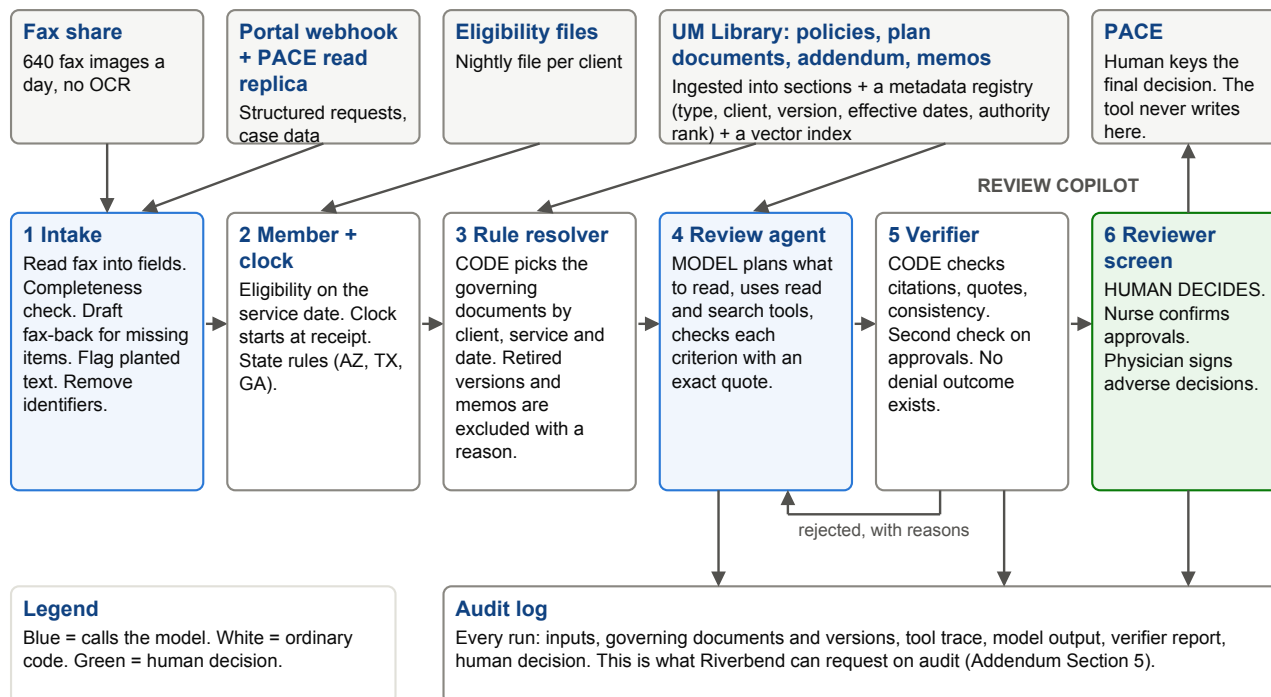
Claim	What the data shows
It is a staffing problem	Nurse count fell 57.7 to 50.6, but on-time rate did not move with it (correlation -0.05) and was already 87% at full staffing. All 64 late Medicare Advantage decisions arrived incomplete. Complete requests are on time 99.8% of the time.
Overturns come from new information at appeal	Complete requests are overturned more often (56.7%) than incomplete ones (48.7%). Denials citing a retired policy are overturned 91% of the time.
It is a wrong-decision problem (Riverbend)	39 of 52 audited errors are rule lookups, not clinical judgement. 34% of 2026 denials cite a retired policy version. A memo kept a 6-week rule after the policy moved to 4 weeks: 72 wrong denials. Kestrel's plan change never reached reviewers: 8 wrong denials.

Diagnosis: late because requests arrive incomplete (122 of 199 hours in late cases is waiting for information), wrong because reviewers cannot find the rule that governs (14 of 38 minutes per case spent hunting). Fix the rule first: it is the only fix that is both faster and more defensible.

Key elements of the build

Agentic RAG with a deterministic rule resolver. Code picks the rule, the model checks the evidence, a person decides.

BELLCOURT SYSTEMS (READ-ONLY). PACE STAYS THE SYSTEM OF RECORD.



- **Registry** of all 38 documents: type, client, version, effective dates, authority rank (GOV-01).
- **Rule resolver in code** picks what is in force on the date of service. The model never chooses a version.
- **Bounded agent** plans its reads; the read tool refuses anything not in force; every Met needs an exact quote.
- **Verifier in code**: citations in force, quotes in the record, outcome consistent, visit arithmetic recomputed. Rejections go back to the model.
- **Second skeptical pass** on every approval.
- **Human decision** with four outcomes and no deny. Decision letter generated from the decision. Priority queue, fax and PDF reading, email.

Governance and security

Built to the Riverbend addendum and the state AI laws

Governance	Security
GOV-01 hierarchy enforced in code: law, then plan document, then policy version by date of service. Memos can never be criteria; excluded documents are shown with the reason.	Sign-in with salted password hashes and signed 8-hour sessions. Roles enforced on the server: only a physician can deny, an auditor is read-only.
No tool may deny, delay or modify a request: there is no deny outcome, requests to providers are drafts a person sends, tests assert it.	Case data and fax images are served only to signed-in users. Supabase holds cases, letters and the audit log.
Audit: every run stores inputs, governing documents and versions, tool trace, output, verifier report and the human decision (addendum s5).	Identifiers (name, DOB, phone, record number, member ID) are removed before the review step. The fax reader runs inside the tenant in production.
Arizona: licensed director signs denials. Texas: AI disclosure in the letter. Georgia: human review before denial. 60-day notice to Riverbend on day 0; pilot on ERISA plans meanwhile.	Provider text is untrusted: planted instructions are flagged, cannot be evidence, and do not change the outcome. Tested with 10 injections and 2 real planted faxes.

How I evaluated it

Bellcourt's own audited cases, the live queue, hostile and broken inputs

117/120

Agrees with the QA auditor

Humans: 68/120. Standard 95%

0

Wrongful approvals, 0 denials issued

by construction and by test

23/24

Adversarial and broken inputs

the 1 failure went to manual review

36/36

Unit tests, no model

resolver, verifier, roles, letters

Where the human reviewer went wrong	Copilot correct
Memo overrode the policy	18 of 18
Retired policy version applied	14 of 14
Client plan rule missed	6 of 7
Clinical facts misread	10 of 10

- **Planted instructions:** 10 of 10 flagged and ignored, plus the two real faxes in the pack.
- **Policy time-travel:** same record, different service date, correct version each time (7 of 7).
- **Missing and wrong inputs:** empty notes, unknown service, no date, unknown member, wrong client: safe outcome, no crash.
- **Honest limits:** results vary slightly between runs (27 to 29 of 30 on the queue), always toward a person. Prompts were tuned on these cases: a fresh audit sample is the next test.

What Bellcourt should do next

Questions

When	What
Day 0	Send Riverbend the 60-day notice. Withdraw the memo. Re-review the affected denials. Report timeliness from receipt.
Days 1 to 30	Host in the Azure tenant. Connect read-only. Load all 38 plan documents. Replay closed cases: 95% on a fresh sample, zero wrongful approvals.
Days 31 to 60	Pilot with 8 senior nurses and 2 intake staff on employer plans.
Days 61 to 90	Riverbend go-live with Arizona routing and Texas disclosure.

Staffing: fill the 7 vacancies, do not add 20. Rule-hunting is 19 nurses' worth of time today. | Live: bellcourt-review-copilot.vercel.app | Code: github.com/devaramanelohith/bellcourt-review-copilot